

Before Lecture 6

- Given the commodities chosen to produced, the firm's goal of decision making is to determine the quantity of each input so that it can maximize its total profit. Notice that the total profit is equal to total revenue (TR) minus TC.
- This goal is closely related to the firm's technology being able to transform inputs to outputs. Its technology will determine its form of production function
- Total profit is related to the level of total production (TP) and the corresponding TC
- To pursue profit maximization, Lecture 5 focuses on the long-run choices of input bundles being able to have lowest LAC given the level of TP

- However, we still cannot obtain the supply curves mentioned in Lecture 2 from the analyses in Lecture 5 that profit maximization results from the choice of input bundles having the lowest LAC. Why?
- Total profit = total revenue – total cost
 - The problem lies in the total revenue. Total revenue is closely related to the level of TP and the price of per unit of commodity.
 - **Price level of the commodity planning to produce will affect the firm's decision on the level of TP, and the level of chosen TP will influence the choice of input bundles having the lowest LAC.**
 - The way how the price level of the commodity influence the decision on the level of TP is closely related to another limitation when the firm makes its decision on the quantity supplied: the competitiveness of the commodity market
- In the following Lecture, we will study how the competitiveness of the commodity market influences the firm's decision on the level of TP and derive the corresponding supply curve. In Lecture 6, we will start from the supply curves implied by the firms in competitive markets

Lecture 6: 完全競爭市場

(Firms in Competitive Markets)

Outline

- 6-1: Commodities and the firms' behaviors of supply in competitive markets
- 6-2: The competitive firm's short-run decision on the level of TP and the derivation of short-run supply curve
- 6-3: Producer surplus
- 6-4: Market equilibrium in the short run
- 6-5: Market equilibrium in the long run

6-1: 產品結構與廠商的供給行為

(Commodities and the firms' behaviors of supply in competitive markets)

- **Difference between individuals and firms when making decisions**

- Individuals: The basic unit of demand behavior in the commodity market. Compared with the whole quantity supplied in a specific commodity market, each individual's quantity demanded for the commodity is so small that he or she can only be a price taker when making consumption.

These individuals make their own decisions on quantity demanded considering only their own utility functions, the given price of the commodity and their income (i.e., their budget constraint).

It implies that each individual's decision will not influence others'. They are independent.

- Firms: The basic unit of supply behavior in the commodity market.
The scale or size of a firm in a specific commodity market could be large. If it large enough, the firm could be a price maker rather than a price taker.

If the number of firms in the same commodity market is high, they will compete each other by lowering the price of the commodity. Once the price is changed due to competition, each firm's decision on TP will also change. That will further influence the decision on LAC.

Thus, the number of firms in a specific commodity market is closely related to those firms' decisions on quantity supplied.

- **Thus in Lectures 6,7, and 8, we focus on commodity markets.**

There are four types of commodity markets categorized according to competitiveness in those markets.

(1) Perfectly competitive market

- [a] Market with many buyers and sellers
- [b] They trade identical products
- [c] Thus, both buyers and sellers are price takers
- [d] Firms (i.e., the sellers) can freely enter or exit the market

(2) Monopoly

- [a] A firm that is the sole seller of a product without close substitutes.
- [b] The firm has market power, the ability to influence the market price of the product it sells
- [c] It arises due to barriers to entry. Other firms cannot enter the market to compete with it.

(3) Monopolistic competition

- [a] Market with many sellers
- [b] Though they trade the same product, they have product differentiation, like food and clothing.
- [c] The sellers are not price takers
- [d] Firms (i.e., the sellers) can freely enter or exit the market

(2) Oligopoly

- [a] Market structure in which only a few sellers offer similar or identical products
- [b] A firm's decisions about price and quantity supplied can affect other firms and cause them to react. The firm will consider these reactions when making decisions.

6-2: The competitive firm's short-run decision on the level of TP and the derivation of short-run supply curve

A competitive market has features as follows:

- (1) Great number of firms (suppliers or sellers)
- (2) They produce identical products
- (3) Any firm in or out of the market can produce the product almost without any difficulties. Thus, firms can freely enter or exit the market
- (4) Firms are price takers. Once one of the firm set a higher price in the competitive market, all of the buyers will choose the identical one provided by the firms offering the lower price
- (5) Thus, no competitive firms have market power to influence the price

- Suppose that the price of the product in the competitive market is P_0 and all of the firms in that market are price takers. Given the price, a firm produce STP units of the product. The firm's total revenue (TR) can be expressed as

$$TR = P_0 \times STP$$

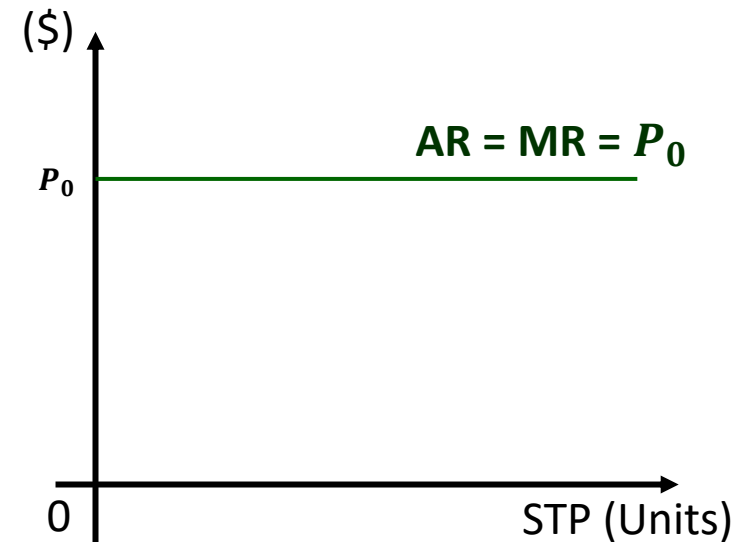
- The average revenue (AR) can then be expressed as

$$AR = \frac{TR}{STP} = \frac{P_0 \times STP}{STP} = P_0$$

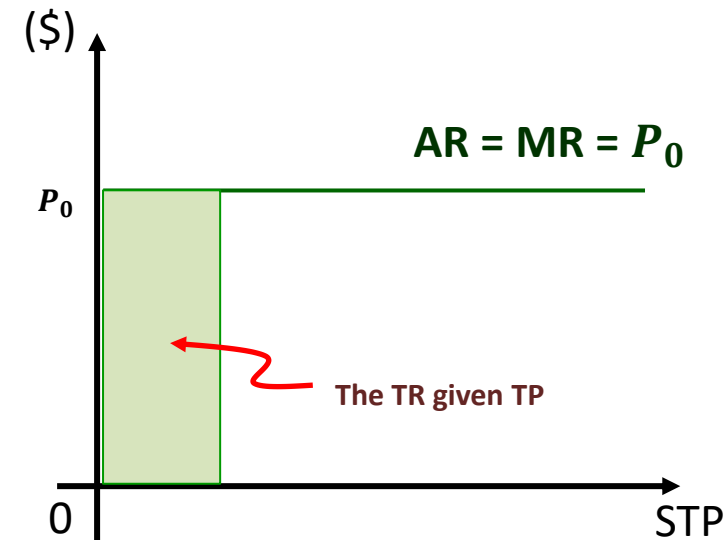
- Because the firm is a price taker, the increase in TR that arises from an extra unit of TP is also P_0 . Thus, the marginal revenue (MR) can be expressed as

$$MR = P_0$$

- Thus in this market, the firm can produce any units of products if it accepts the market price P_0 . However, it needs take production cost into account. Given the price, the firm will choose the TP being able to maximize its economic profits.



- Competitive firms have horizontal curves of AR
- AR and MR are indifferent to TP because all of the competitive firms are price takers



- Recall that in **Lecture 5**, in the short run, the firm can only adjust the quantity of variable factor inputs. Given the TP, the firm has the corresponding SAVC, SAFC, and SAC.
 - The SAC curve is U-shaped, and the SMC curve crosses the lowest point of the SAC and AVC curve

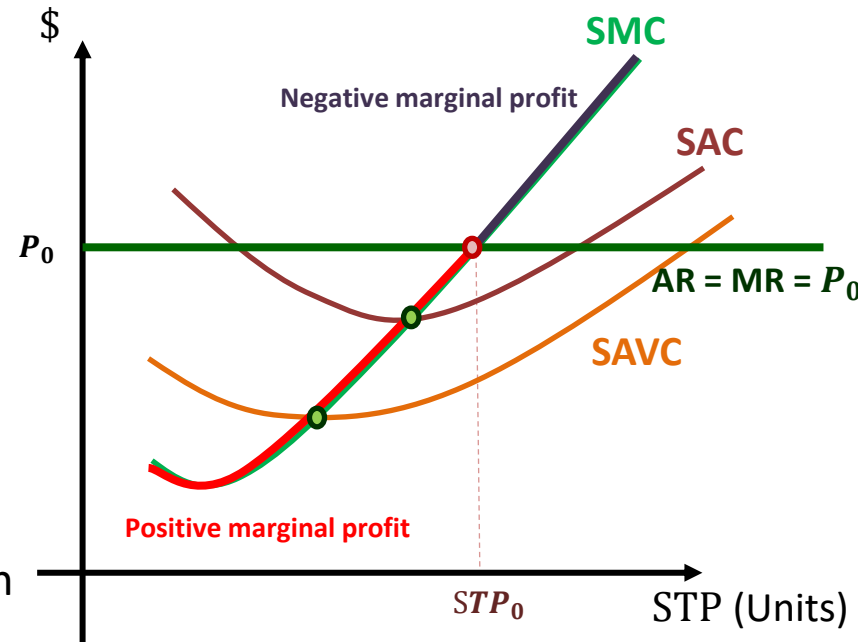
- Recall:**

SMC : an increase in STC that arises from an extra unit of STP

MR: an increase in TR that arises from an extra unit of STP

Thus, a competitive firm will increase production if $MR > SMC$ because the marginal profit is positive; it will stop producing if $MR < SMC$ because the marginal profit is negative

- Thus, a profit maximization competitive firm will let $STP = STP_0$ such that $MR = SMC = P_0$ when the price of its product is P_0

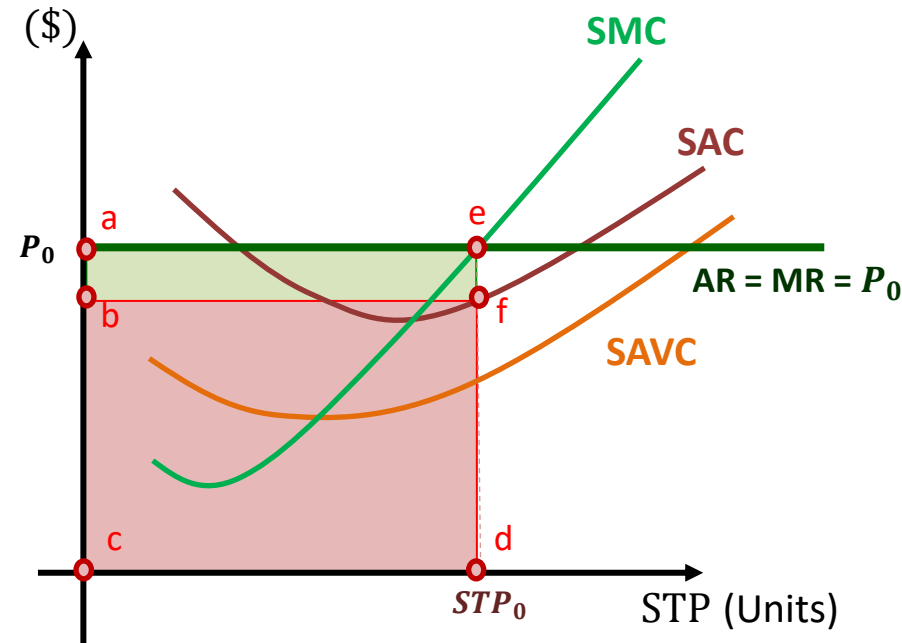


Think:

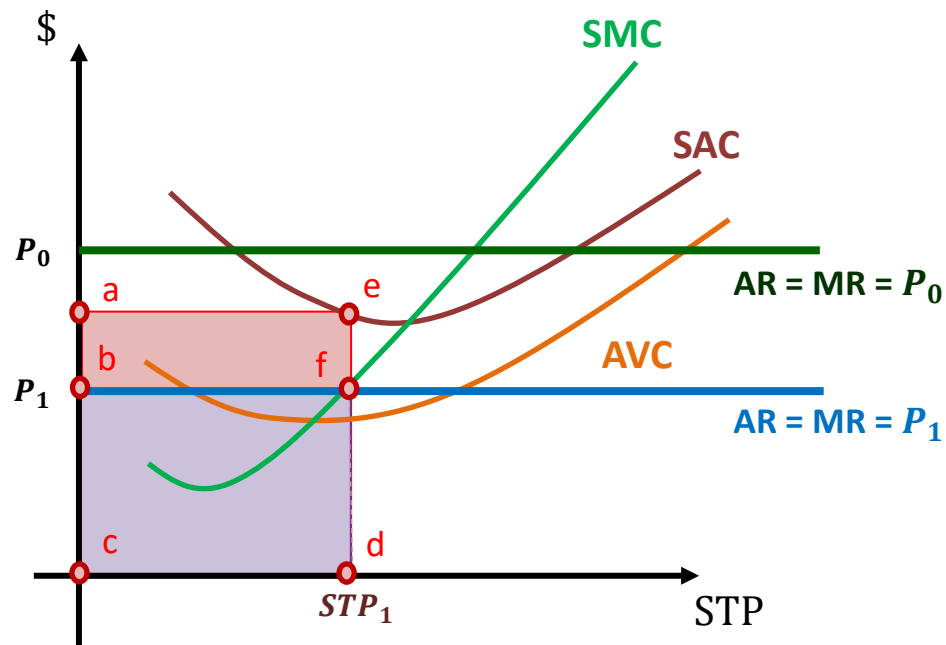
Total profit = TR – TC

- Maximal total profits can be achieved through maximal TR and minimal TC
- However, could the maximal total profit implied by $P_0 = MR = SMC$ refers to maximal TR and minimal TC?

- Given the $TP = STP_0$. The length of $\overline{ed}$ indicates the corresponding AR, and the length of $\overline{fd}$ indicates the corresponding SAC. Thus, the length of $\overline{ef}$ indicates the profit earned per unit of production.
- Due to the fact that $TP = STP_0$ and the price of per unit of product is P_0 , the region acde indicates TR.
- And the corresponding STC is the region bcdf
- Thus, the **short-run total profit** is the region abef if the $STP = STP_0$, and the profit is positive ($\because$ the region acde $>$ bcdf)



- When the price of the product is P_1 , the firm will let $TP = TP_1$ so that $MR = SMC = P_1$. That will maximize the firm's total profit.
- Due to the fact that $TP = TP_1$ and the price of per unit of product is P_1 , the firm's TC is the region acde
- And the corresponding TR is the region bcdf.
- Thus, even the firm let $TP = TP_1$ to maximize its profit when the price is P_1 , the short-run total profit is still negative ($\because$ the region bcdf $<$ acde).



- Recall:

$$\therefore STC = STFC + STVC$$

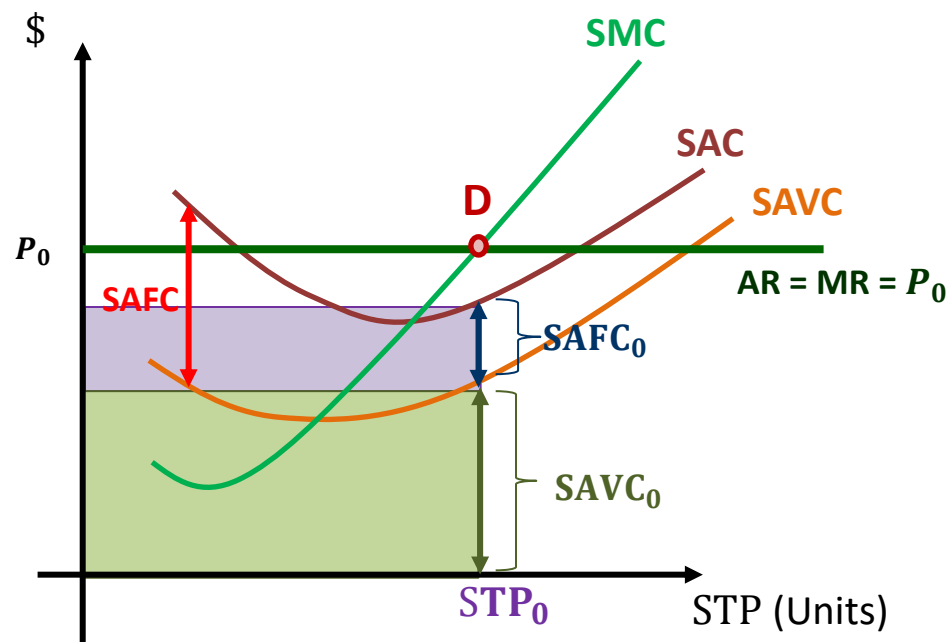
$$\therefore STC/STP = (STFC + STVC)/STP$$

$$\rightarrow SAC = SAFC + SAVC$$

$$\therefore SAFC = SAC - SAVC$$

- Notice:

TFC is the cost paid once the firm plans to produce. Thus, in the short run, it is indifferent to STP. Because of this feature, the SAFC decreases with STP.



- When $MR = SMC = P_0$, the MR line and SMC curve intersect at D, and the corresponding optimal STP is STP_0 . When $STP = STP_0$, the corresponding SAVC is $SAVC_0$ and AFC is AFC_0 . Thus,

$$\text{total variable cost } STVC_0 = STP_0 \times SAVC_0$$

(the green region)

$$\text{total fixed cost } STFC_0 = STP_0 \times SAFC_0$$

(the purple region)

$$\text{total revenue } TR_0 = STP_0 \times P_0$$

- When does a competitive firm stop producing ?
Consider three market price: $P_0 > P_1 > P_2$

Case 1: when the market price is P_0 :

Let $MR = SMC = P_0$. The MR line and SMC curve intersect at D and the corresponding profit maximizing STP is STP_0 .

Then,

$TR_0 = STP_0 \times P_0$: region ABCD

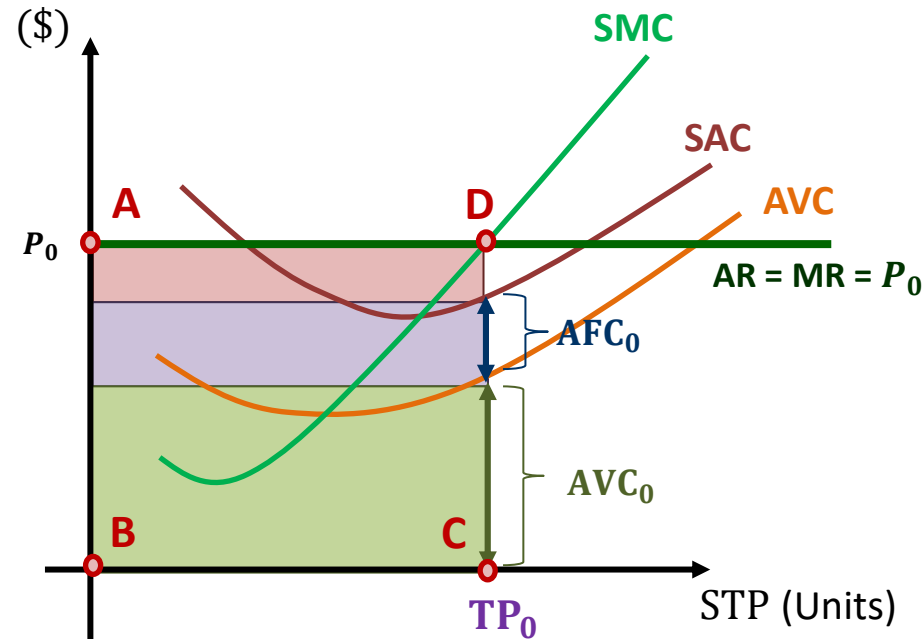
$STVC_0 = STP_0 \times SAVC_0$: green region

$STFC_0 = STP_0 \times SAFC_0$: purple region

Because

region ABCD > green + purple region

The total profit is positive (red region), so the firm will keep producing when price is P_0



Case 2: when the market price is P_1 :

Let $MR = SMC = P_1$. The MR line and SMC curve intersect at D and the corresponding profit maximizing STP is STP_1 .

Then,

$TR_1 = STP_1 \times P_1$: region ABCD

$STVC_1 = STP_1 \times SAVC_1$: green region

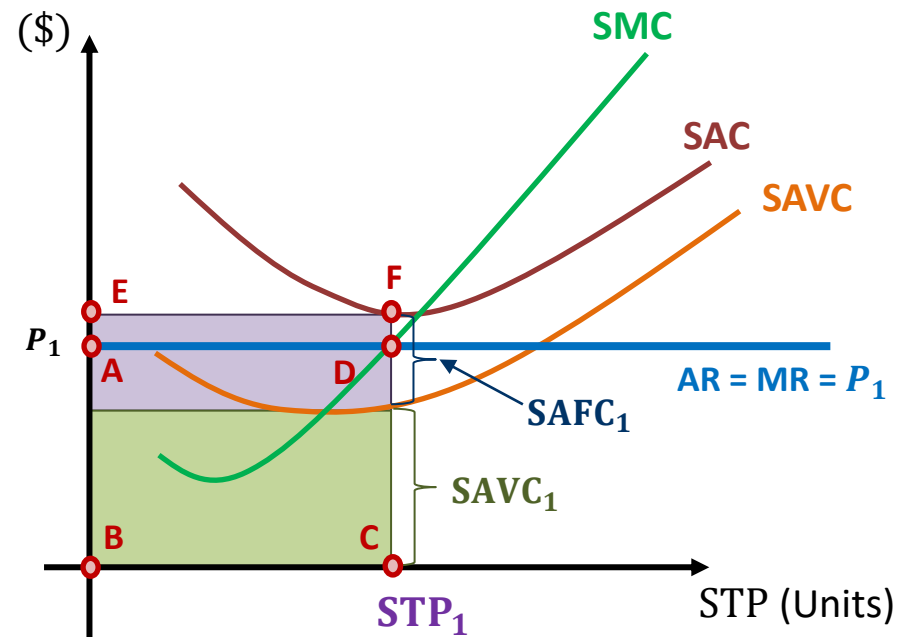
$STFC_1 = STP_1 \times SAFC_1$: purple region

Because

region ABCD < green + purple region

The total profit is negative, the this loss can be illustrated through the region EADF ($STVC_1 + STFC_1 - TR_1$)

- If the firm stop producing, it needs pay $STFC_1$ because this is the cost once the firm plan to manufacture the product. Thus, if the firm stops producing when the market price falls from P_0 to P_1 , its loss is $STFC_1$, the purple region.



- Due to the fact that purple region > region EADF the firm will keep producing because the loss is smaller than that met when stop producing.

Case 3: when the market price is P_2 :

Let $MR = SMC = P_2$. The MR line and SMC curve intersect at D and the corresponding profit maximizing STP is STP_1 .

Thus,

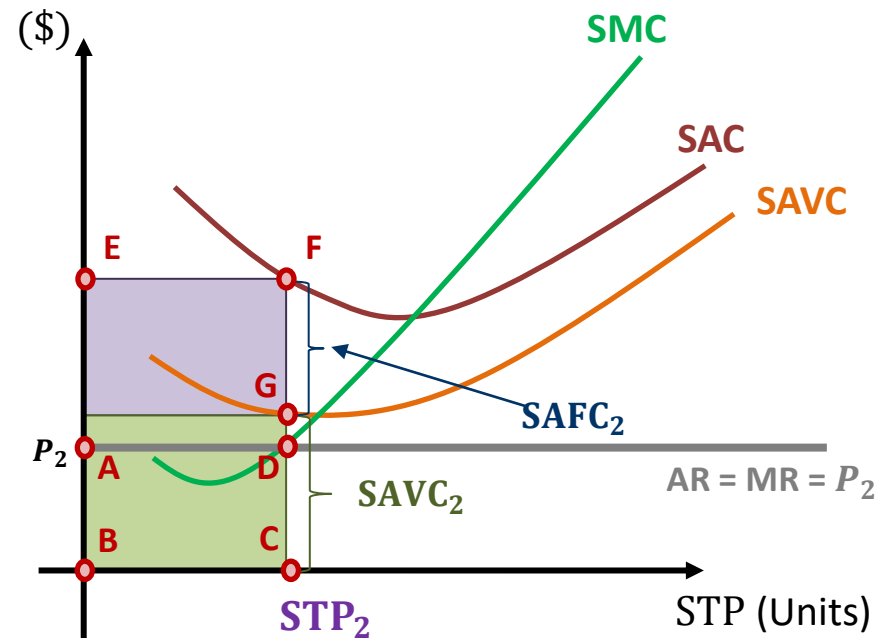
$TR_2 = STP_2 \times P_2$: region ABCD

$STVC_2 = STP_2 \times SAVC_2$: green region
($SAVC_2$ is $\overline{CG}$)

$STFC_2 = STP_2 \times SAFC_2$: purple region
($SAFC_2$ is $\overline{FG}$)

It can be observed that
region ABCD < green + purple region
and
region ABCD < green region.

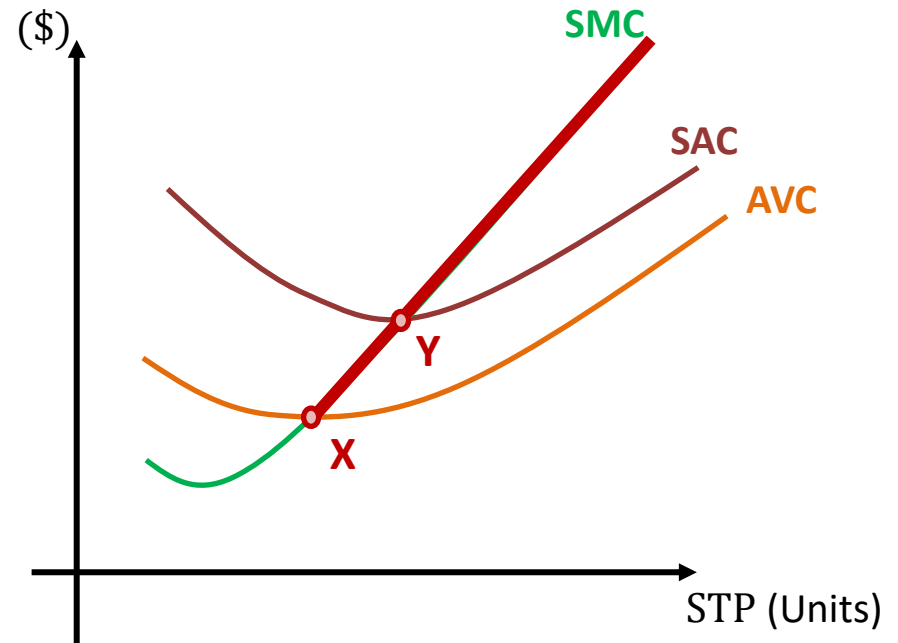
Thus, if the firm let $STP = STP_2$ when the price is P_2 , the total profit is negative and the loss can be illustrated through region EADF



- If the firm stop producing, it will lose TFC_2 , the purple region.
- Due to the fact that
region EADF > purple region,
the firm will stop producing when
the price falls from P_0 to P_2 .

Brief Summary

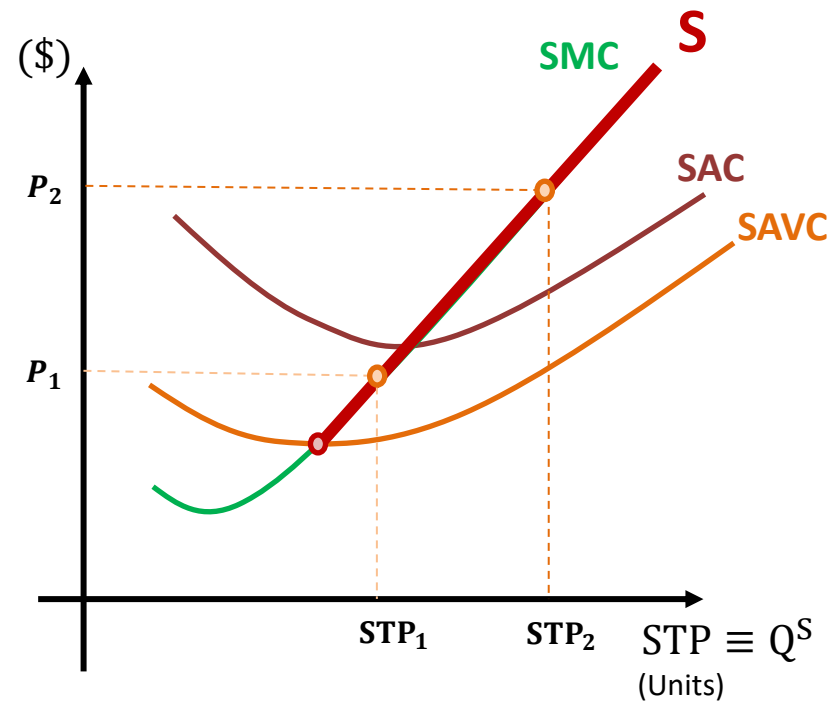
- Firms in competitive markets are price takers. Thus when the price of the commodity they produce is P , their $AR = MR = P$.
- In the short run, those firms will choose their TP so that the corresponding $MR = P = SMC$
- If $MR = SMC = P > SAVC$, firms will keep producing even though their total profits are negative (Case 1 and 2)
- If $MR = SMC = P < SAVC$, firms will stop producing, because the losses met as they stop producing is less than those as they keep producing (Case 3)



- Points X and Y are lowest point on the SAVC and SAC curve.
- Thus in the short run, the supply curve implied by a firm in a perfectly competitive market is the part of the SMC curve that $SMC > SAVC$.**

6-3 Producer Surplus

- The STP on the X axis can be denoted by Q^S , and the Y axis indicates the price of per unit product, P . Then the red-colored S curve is (short-run) supply curve as in Lecture 2.
- The price on the Y axis represents the minimal price the firm is willing to offer given STP.



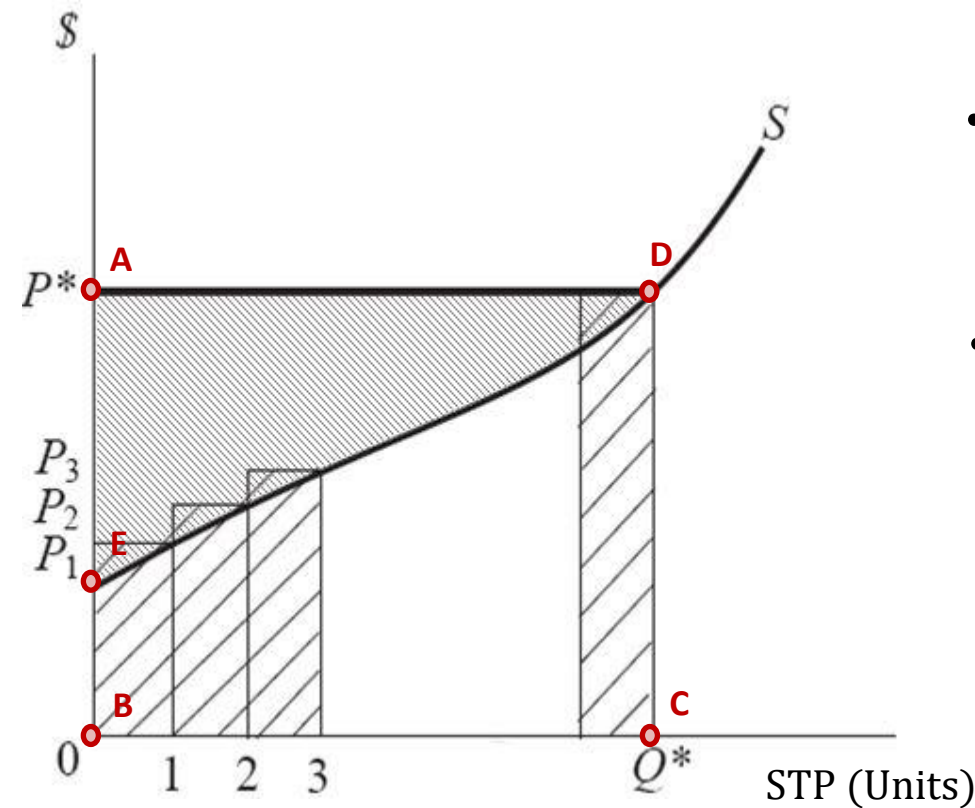
- If a firm determine to let its $TP = Q^*$, then its corresponding lowest price it willing to offer is P^* . Let the market price of per unit of product is also P^* .

- Then the firm's total revenue is

$$TR = P^* \times Q^*$$

- If the firm supplies Q^* units of products, the minimal total revenue it is willing to accept is the region of the summation of all lash rectangles (region EBCD). However, the total revenue it actually receives is $P^* \times Q^*$, the region ABCD.

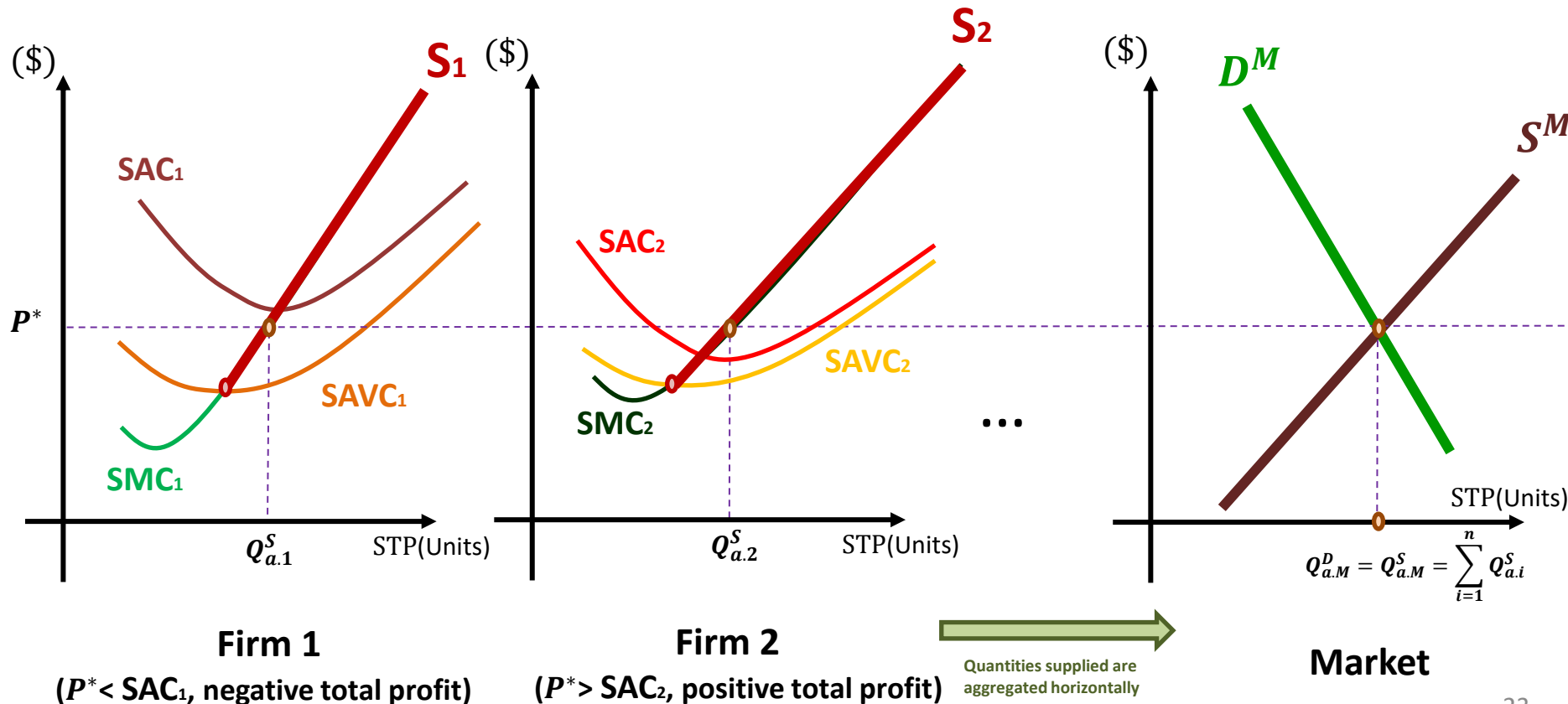
- The region AED is called producer surplus. A producer surplus is a difference between how much the producer receives and how much of a good the producer is willing to supply



6-4 Market equilibrium in the short run

- According to the definition in Lecture 2, we know that market equilibrium is a status that product price has reached the level where quantity supplied equals quantity demanded.
- According to Lecture 4, we know that buyers facing budget constraints make the consumption decisions being able to maximize their own utility. With the decisions, we get the market demand curve for a given product.
- According to Lecture 6-2, we derive the supply curve for each individual firm (supplier). We then can obtain the market supply curve by summing all of the individual supply curves horizontally.

- Suppose that the price of per unit of product a is P^* .
- Both of the buyers and the firms in the competitive market are price takers
- Suppose that there are n firms producing product a. They are all the firms maximizing their own profits.
- In the short run, market of product a achieve equilibrium through horizontal aggregation of quantities demanded and supplied. Then we have equilibrium price P^* . With this price, firms might have positive or negative total profits



6-5 Market equilibrium in the long run

- Firms can make two kinds of adjustments from short run to long run
 - (1) They make decision on whether they enter the target market or leave the market they stay.
 - (2) If they choose to stay, they will adjust their firm size (the quantity of production factor input) to keep profitable.

- Adjustment of whether to enter or leave:

- Suppose that the market of product b is a perfectly competitive market and there are n firms in the markets. The n is very large. If the market price of the product is P^Δ , the total profit of the firm i could be expressed as

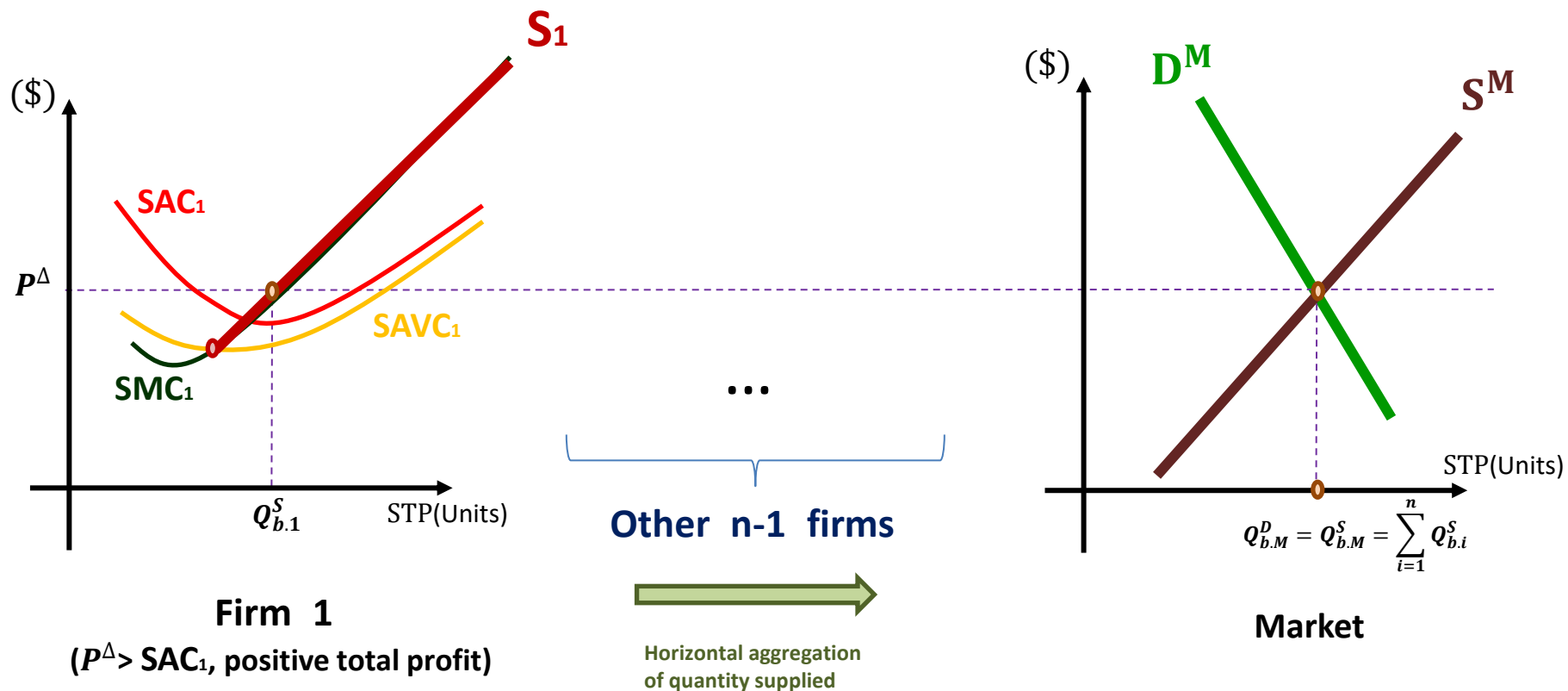
$$(P^\Delta - SAC_i) \times STP_i$$

- Firms may have different cost structures (i.e., SAVC, SAC and SMC). Thus, given the market price P^Δ , some firms may have positive total profits, but some have negative. Notice that the firms staying in the market have

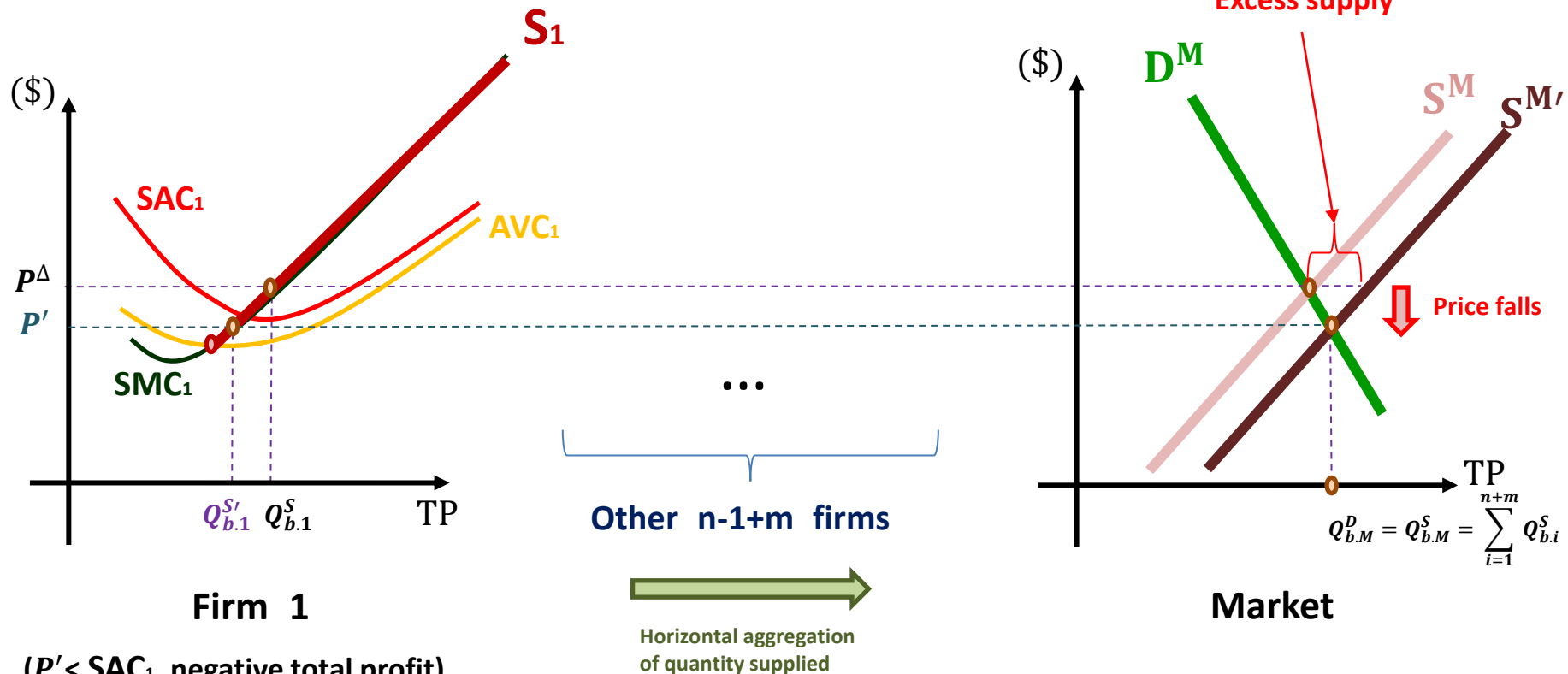
$$MR_i = SMC_i = P^\Delta > SAVC_i$$

in the short run.

- For example, the profit-maximized firm 1 will let $STP = STP_1$ when the market price is P^Δ . It has positive total profits because $P^\Delta > SAC_1$.
- Because of the fact that competitive firms can freely enter and leave the market, the total number of firms will increase once there are producers having positive total profits.
- The relation between market demand curve and the supply curve implied by the firm 1 can be illustrated as follows:

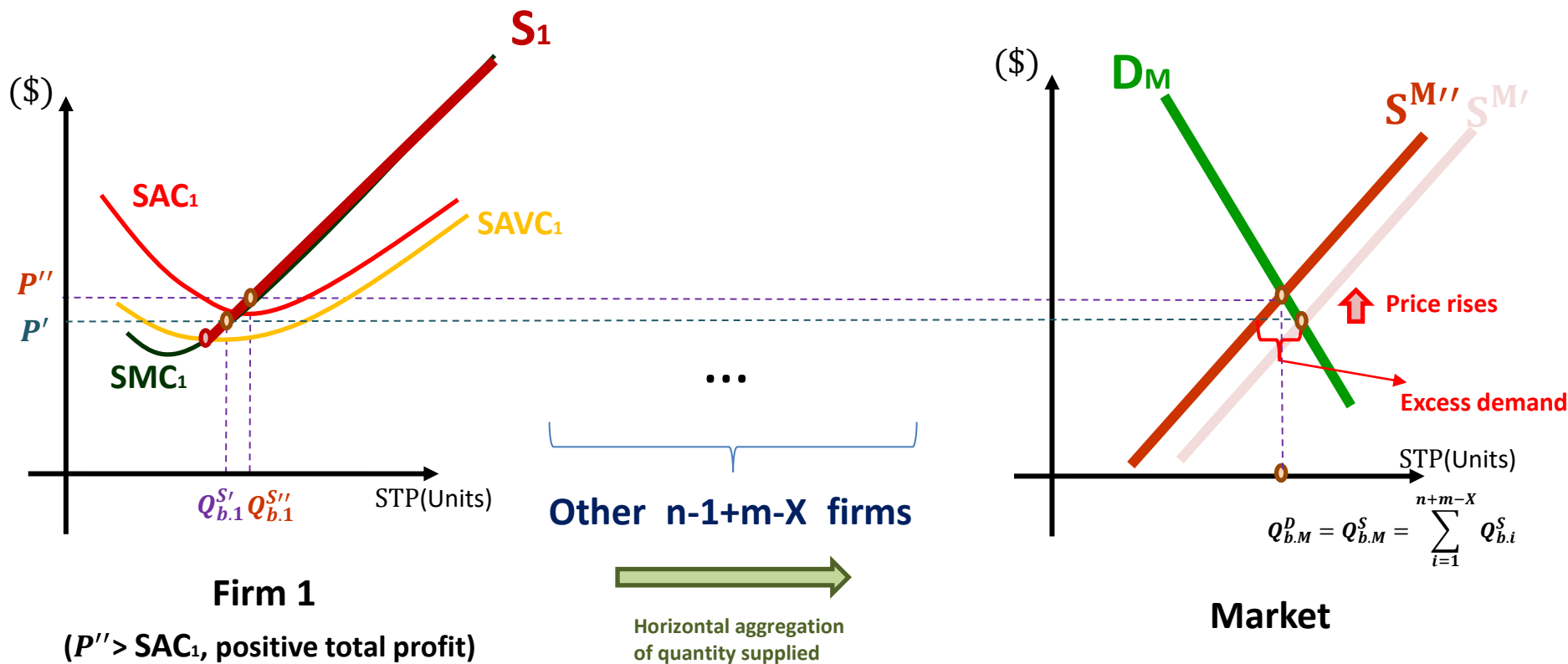


- Once the firm 1 has positive total profit, other firms will have incentives to enter this market. Suppose that m firms decide to enter this market. If m is great enough, that will push the market supply curve S^M to the right (see the next slide).



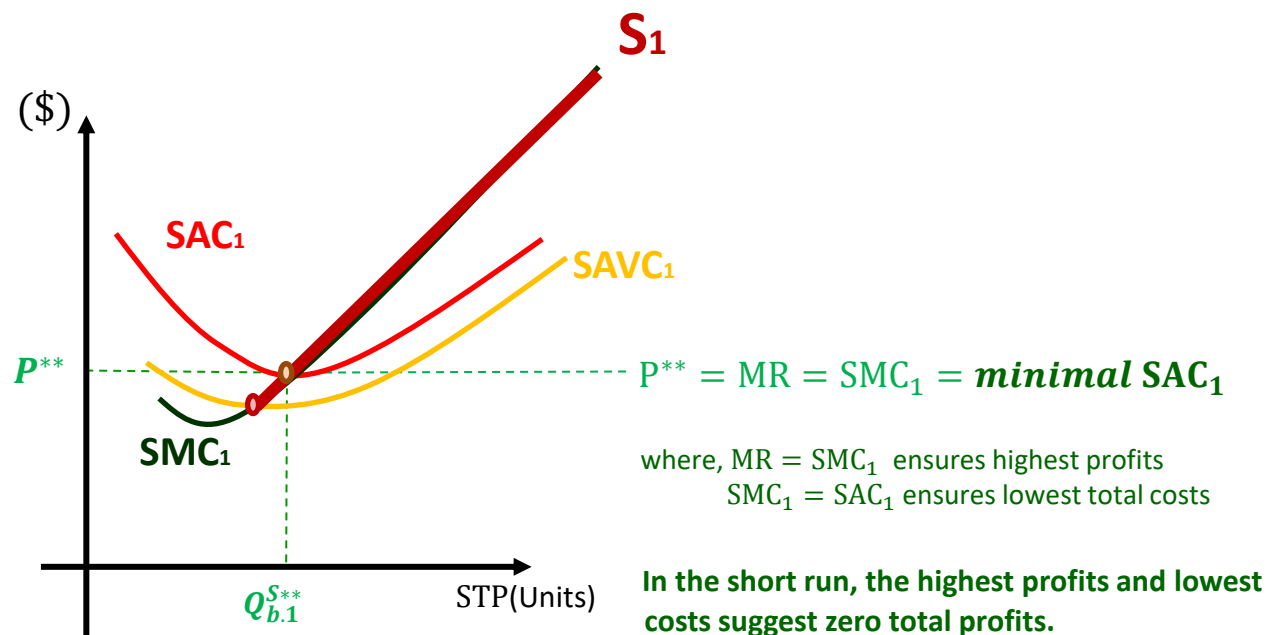
$(P' < SAC_1, \text{negative total profit})$
 $(P' > SAVC_1, \text{keep staying in the market})$

- Given that the price of the product and the market demand curve are unchanged, the rise in quantity supplied causes excess supply. The price will fall from P^Δ to P' to push the market back to the equilibrium.
- The new price will make the firm 1 to have negative total profit, but it will still stay in the market because $P' > SAVC_1$. However, this new price may make other firm exit the market (**those firm having $P' < SAVC_i$**).



- Suppose that X firms decide to exit the market when the price of the product is P' . If the X is great enough, that will push the market supply curve to the left. Other things being equal, the market will have excess demand, and then the price will rise from P' to P'' to push the market back to the equilibrium.
- Given the higher price P'' , the firm 1's total profit changes from negative to positive. Once the firm 1 has a positive total profit, other firms out of the market will have incentives to enter this market. Then same procedure goes again.

- Therefore, the same procedure will go on because in the competitive market firms can freely enter and leave the market. The procedure will stop once the market price is P^{**} such that $P^{**} = SAC_1$. This price implies zero total profits.

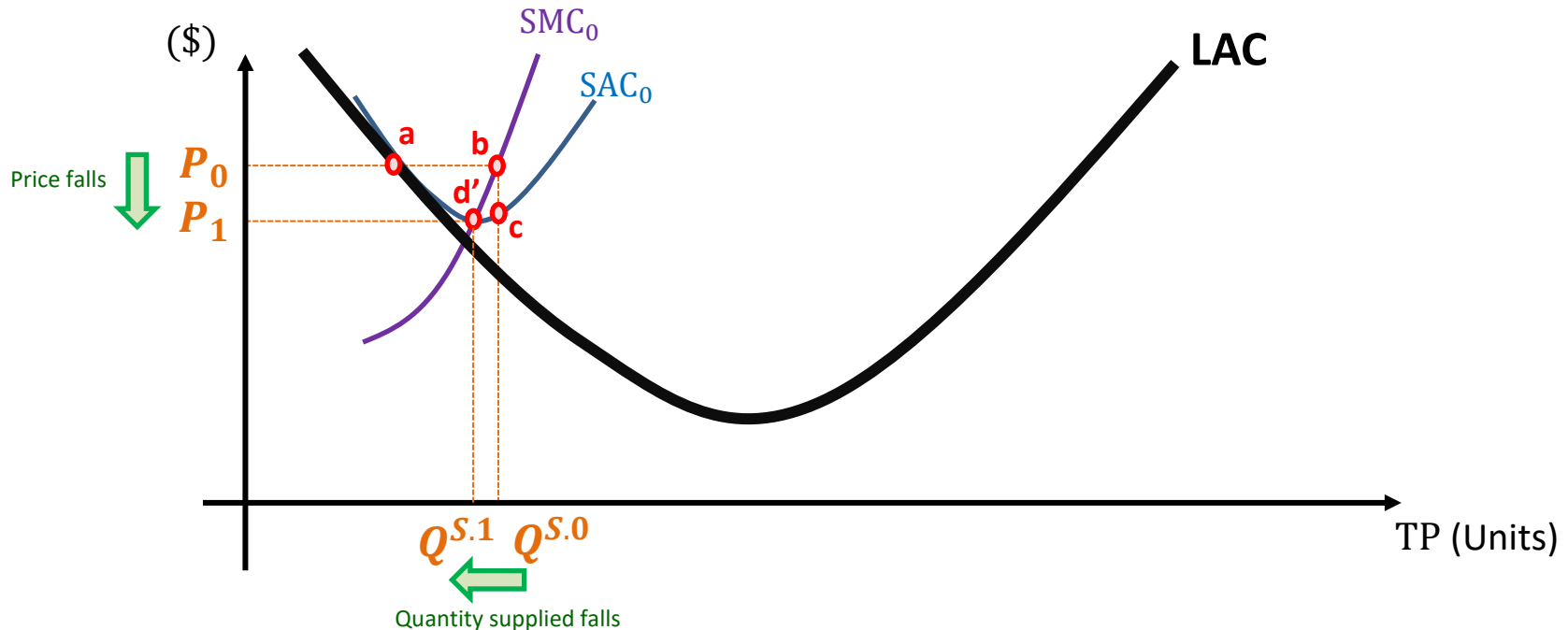


Firm 1

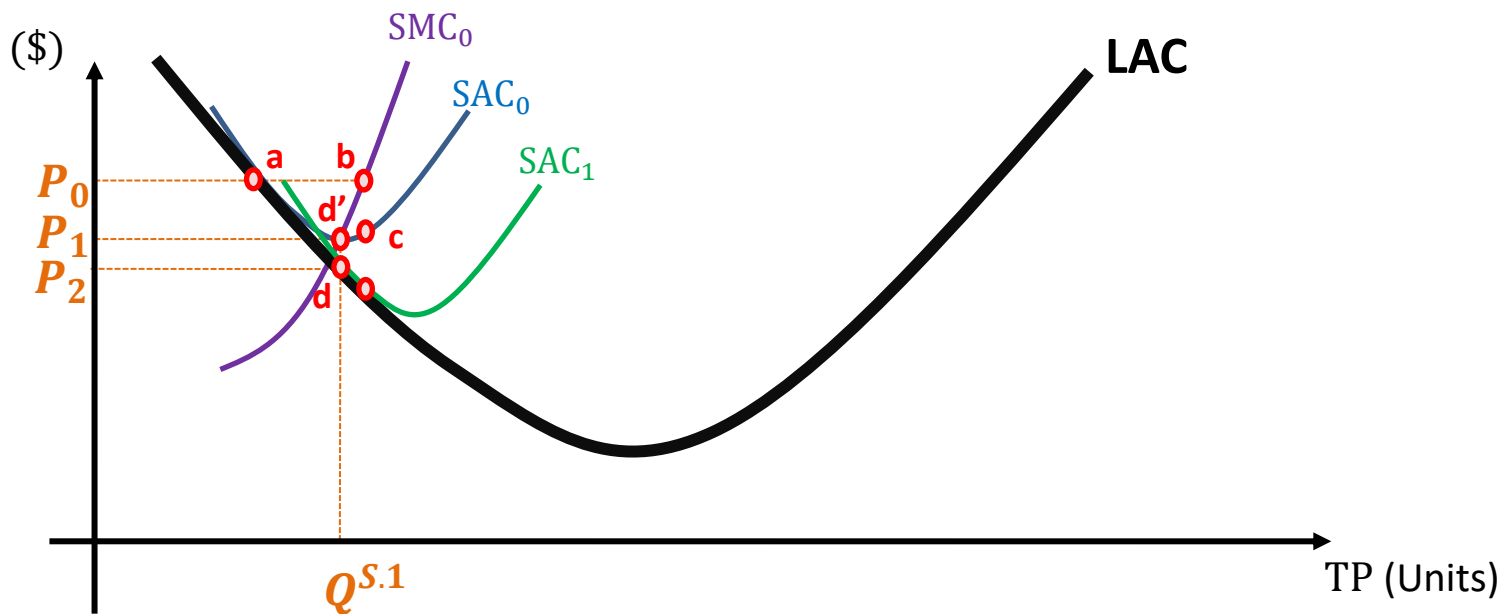
- Why do the firm stay in the market when its total profit is zero in the short run? See the next slide.

- The adjustment of firm size (illustrated by SAC) :

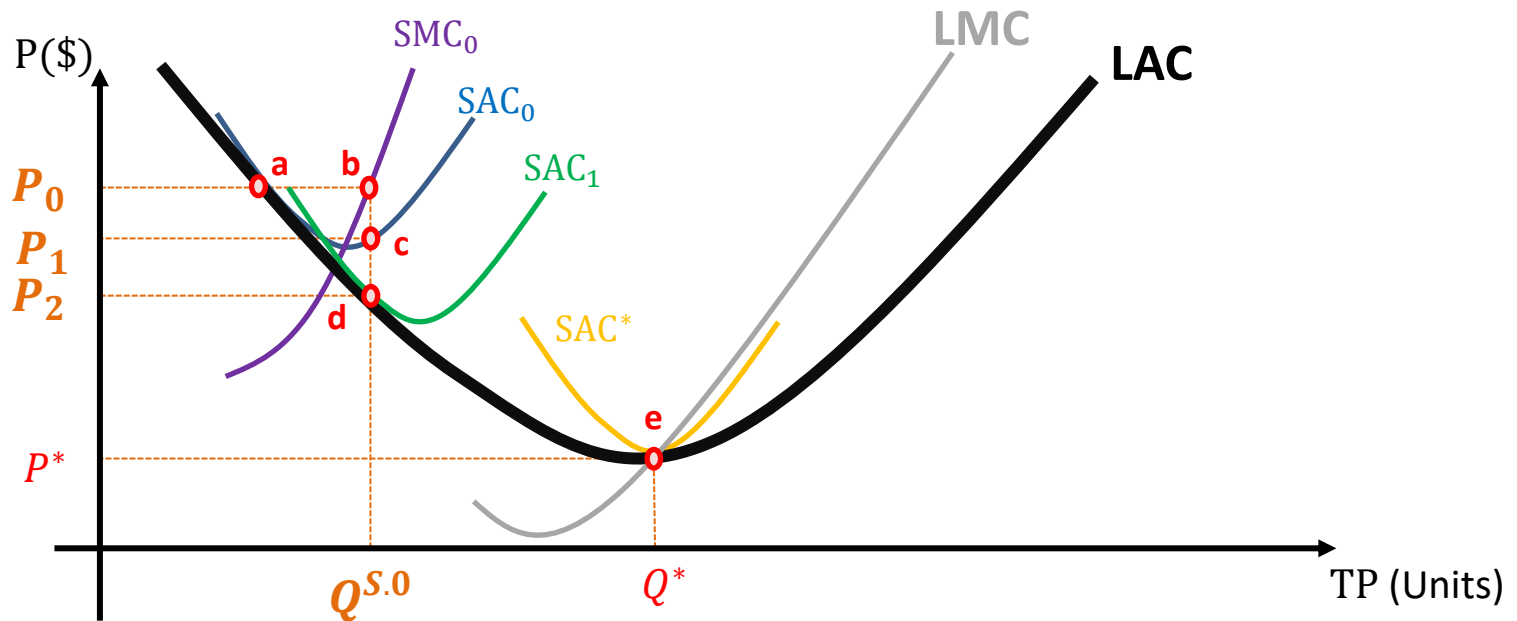
- Recall that the LAC curve is the envelop curve comprised by SAC curves.



- When the market price is P_0 , the optimal size is SAC_0 (according the point a on the LAC curve; a is simultaneously on the SAC_0 and LAC curve). With this size, a competitive firm will decide to let $TP = Q^{S.0}$ because this decision follows $MR_0 = P_0 =$ point b on SMC_0 . Due to the fact that $\overline{bc} > 0$, the firm has positive profit per unit of production (thus positive total profit).
- Because the firm has positive total profit, other firms out of the market will enter. If the number of firms rises and the market demand curve is unchanged, the price falls. If the price falls from P_0 to P_1 , the firm has zero total profit ($\because$ point d' is on SAC_0 curve) if it still choose the size SAC_0 .



- The firm may not leave the market even when its total profit is zero. It can adjust its size.
- When $TP = Q^{S.1}$, the point d tells the firm that it can adjust its size from SAC_0 to SAC_1 to let its total profit positive ($\because \overline{dd'} > 0$) given that the price is P_1 .
- Once the firm adjusting its size to SAC_1 has positive total profit, other firms out of the market will enter to push the quantity supplied of the whole market up. The price of the product will fall further. Then the adjustment procedure mentioned above goes again. The firm size is adjust from small to big, and the price will fall further until.....



- When the price of the product fall to P^* , the firm will have zero total profit because the point e is the lowest point of SAC^* as well as the LAC curve such that

$$P^* = LAC$$

- If all of the competitive firms in the market are price takers, then $MR = P^*$, and the corresponding optimal TP is Q^* such that $MR = P^* = LMC$.
Therefore we obtain the conclusion that in the long run, the firm in the competitive market will let $TP = Q^*$ such that

$$LAC = LMC = P^*$$

- In the long run, the optimal TP for firms in competitive markets is the one such that

$$P^* = LMC = LAC.$$

That is, it is the TP where its LMC and LAC curve intersect (the point e in previous slides).

- This is the results that firms can freely enter or leave market. Firms in the markets finally have zero total profits, because once one of them has positive total profits, others will enter the markets to dilute the profits. In other words, to keep zero profits, firms in the markets need choose the firm size being able to have lowest LAC. We call this “having production efficiency.”
- Thus in the long run, competitiveness will make the price of the product fall, but it will increase the production efficiency.